MILITARY SPECIFICATION

HEADSET-MICROPHONE KIT MK-896A/AIC
(components of)

This specification is approved for use by Communications Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Headset-Microphone Kit MK-896A/AIC. The kit is designed to fit both the regular and extra large size Helmet, Flying Type SPH-4.

1.2 Each MK-896A/AIC is composed of the following components:

1 each Microphone, Dynamic M-87A/AIC
Note: Army version of US Air Force M-87/AIC which basically has immersion requirement instead of explosive decompression requirement. All other parameters are identical and, except for the above mentioned difference, the microphones are interchangeable.
1 each Cable Assembly, Microphone, one end terminated in Plug U-173/U and opposite end connected to the microphone
1 each Boom Assembly, Microphone
1 each Swivel Assembly

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Communications and Electronics Materiel Readiness Command, ATTN: DRDCO-CM-DM, Fort Monmouth, New Jersey 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1 each Earphone H-143A/AIC
Note: Army version of US Air Force H-143/AIC which basically has immersion requirement instead of explosive decompression requirement. All other parameters are identical and, except for the above mentioned difference, the earphones are interchangeable.
1 each Cord Assembly Kit, one end terminated in Plug U-174/U and opposite end branched to attach the Earphones and Connector U-172/U
1 each Jack Holder Plate
Necessary grommets and hardware for attaching the kit to the helmet.

2. APPLICABLE DOCUMENTS

2.1 Issues of Documents. The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	Pallet, Material-Handling, Wood (General construction requirements)
QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-566	Box, Folding, Paperboard
PPP-B-585	Boxes, Wood, Wireboard
PPP-B-601	Boxes, Wood, Cleating
PPP-B-621	Boxes, Wood, Nailed, and Lock-Corner
PPP-B-636	Boxes, Fiber, Solid
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (container grade) and Cut Shapes
PPP-P-291	Paperboard, Wrapping Cushioning
PPP-T-45	Tape Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (for carton sealing)
PPP-T-97	Tape, Pressure-Sensitive Adhesive Filament Reinforced

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MIL-P-116	Preservation Packaging, Methods of
MIL-I-4997	Insulating and Jacketing Compounds for use in Cords, Cordages and Cables
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-C-5898	Cord, Electrical
MIL-C-10392	Cord, Electrical (Audio, Miniature)
MIL-P-11268	Parts, Material and Processes Used in Electronic Communication Equipment
MIL-M-13231	Marking of Electronic Items
MIL-E-25670 (US Air Force)	Earphone, H-143/AIC
MIL-M-26542 (US Air Force)	Microphone, Dynamic, M-87/AIC
MIL-B-43014	Boxes, Water Resistant, Paperboard, Folding, Set-up and Metal-Stayed
MIL-C-45662	Calibration System Requirements
MIL-C-55668(EL)	Cord, Electrical: Audio, Retractile, Subminiature

STANDARDS

FEDERAL

FED-STD-228	Federal Test Method Standard

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners, or Pallet Type Base
MIL-STD-202	Test Methods for Electronic and Electrical Components Parts
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461 (Notice 4(EL))	Electromagnetic Interference Characteristics, Requirements for Equipment

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MIL-STD-462 (Notice 3(EL))	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810	Environmental Test Methods

DRAWINGS

NAVY

SK-N-864	Simulated Gun Blast Producing Equipment

Copies of specifications, drawings, standards, and publications required by suppliers in connection with specific procurements should be obtained from the procuring activity or as directed by the Contracting Officer. Both the title and number or symbol should be stipulated when requesting copies.

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Acoustical Society of America (ASA)

ANSI	Z24.9-1949	Coupler Calibration of Earphones
ANSI	S1.10-1966	USA Standard Method for the Calibration of Microphones

(Application for copies should be addressed to the Acoustical Society of America, Standards Manager, 335 East 45th St., New York, New York 10017.)

American Society of Testing and Materials

B33-63	Standard Specification for Tinned Soft or Annealed Copper Wire for Electrical Purposes

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

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3. REQUIREMENTS

3.1 Description. Headset-Microphone Kit MK-896A/AIC consists of the component parts listed in 1.2. The dynamic earphones are waterproof and blastproof. The dynamic, noise cancelling microphone and boom assembly is waterproof and blastproof. Headset-Microphone Kit MK-896A/AIC shall fit both the regular and extra large size Helmet, Flying Type SPH-4.

3.1.1 General. The requirements of MIL-E-5400 for Class 1 equipment apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.1.1.1 Approval of Non-Standard Parts and Material. The requirements of MIL-E-5400 for Category III equipment shall apply.

3.1.1.2 Standard Model. When stated in the contract or purchase order, a model equipment, MK-896A/AIC, will be made available to the contractor, on loan, to be used as a general guide, except in instances where specific improvements are detailed herein or where requirements dictate changes. The model equipment shall not be subjected to any destructive tests nor shall any part or component (with the exception of subassemblies retained by machine screws or the like) be removed or disassembled.

3.2 Parts, Materials and Processes; General. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply.

3.2.1 Magnetized Materials. The materials used in the magnetic circuit shall be of such a character and shall be so processed and assembled that the microphone and earphone elements will not suffer objectional degradation in performance due to loss in magnetization over long periods of storage or service.

3.2.2 Marking. Marking shall conform to MIL-M-13231.

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3.2.2.1 Serial Numbers. Serial numbers are not required.

3.3 Construction. The construction of parts for the assembly of the Headset-Microphone Kit MK-896A/AIC shall conform to the requirements specified herein and shall be suitable for extended service use in military aircraft.

a. Moisture Barrier Seal. A moisture barrier shall be provided as part of microphone. The moisture barrier shall serve both as a breathblast shield and a shield to prevent entrance of moisture into the microphone.

b. Moisture Barrier Guard. The moisture barrier guard shall be so constructed that it can be easily replaced by another moisture barrier guard without the use of any special mechanical equipment.

c. Microphone Positioning Adjustment. The microphone positioning adjustment shall provide a non-critical adjustment to obtain a torque of 30 to 50 ounce inches. From an initial setting of 40 ounce inches, the torque shall not change in excess of 20 percent after subjection to any or all of the environmental tests, including the endurance test for the microphone adjustment bearing specified herein.

3.4 Microphone.

3.4.1 Microphone Response. The frequency response of the microphone with the moisture barrier guard in its place shall be essentially flat from 300 to 5000 Hz. The minimum power output of the microphone shall be -64 dBm (.00025V) at 1000 Hz when measured in accordance with 4.7.1. In addition, the response shall fall within the envelope of Figure 1. During this test, the microphone shall be terminated with a non-inductive load of 5 ohms.

3.4.2 Distortion. Total harmonic distortion shall not exceed 6 percent at a sound pressure level of 125dB referenced at 20 micropascals over the frequency range of 700 to 5000 Hz. (See 4.7.2.)

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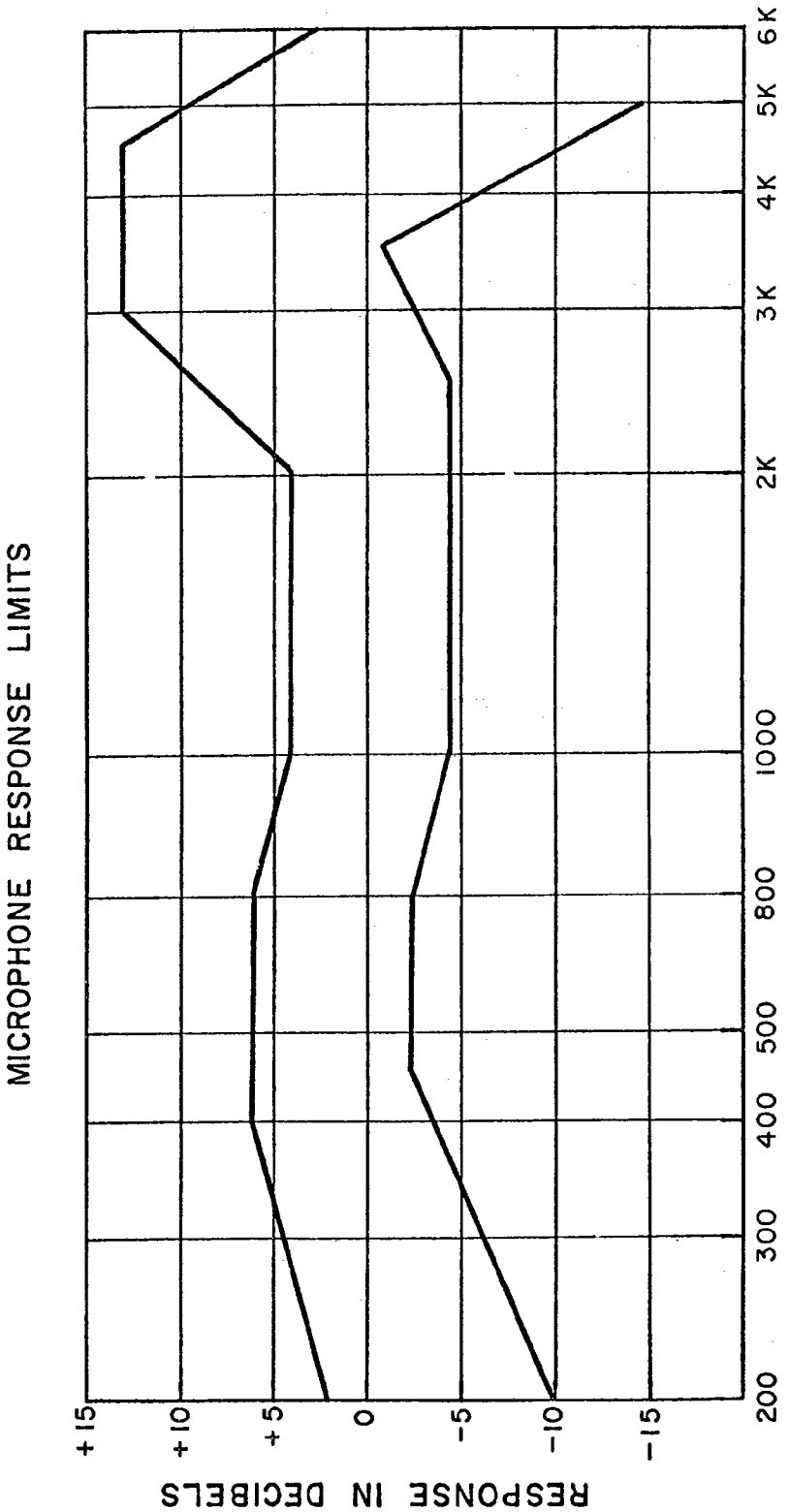


FIGURE I

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3.4.3 Signal-to-Noise Ratio. The signal-to-noise ratio of the microphone shall be at least 18dB (signal over noise) when measured in accordance with 4.7.3.

3.4.4 Dielectric Strength, and Insulation Resistance. There shall be no evidence of insulation breakdown when the microphone is subjected to a voltage of 100 VAC 60 Hz for ten (10) seconds applied between the terminals of the microphone and the insulated outer metal parts of the microphone. There shall be no decrease in insulation resistance below 1 megohm following the above test. (See 4.7.4.)

3.4.5 Impedance. The impedance of the microphone at 1000 Hz shall be within the limits of 3 ohms and 7 ohms. (See 4.7.5.)

3.4.6 Microphone Moisture Barrier. The microphone shall be equipped with a moisture barrier for protection against icing, dust and wind. The specified response curve and output level of the microphone shall be measured with the moisture barrier guard in place on the microphone.

3.5 Earphone.

3.5.1 Earphone Response (See 4.8.1). The output of the earphone shall be not less than 102dB above a reference level of 20 micropascals when 1 milliwatt rms power at 1000 Hz is applied to the earphone terminals. The response of the earphone at any frequency shall not deviate from 1000 Hz response by more than the values shown in Table I. (See Figure 2)

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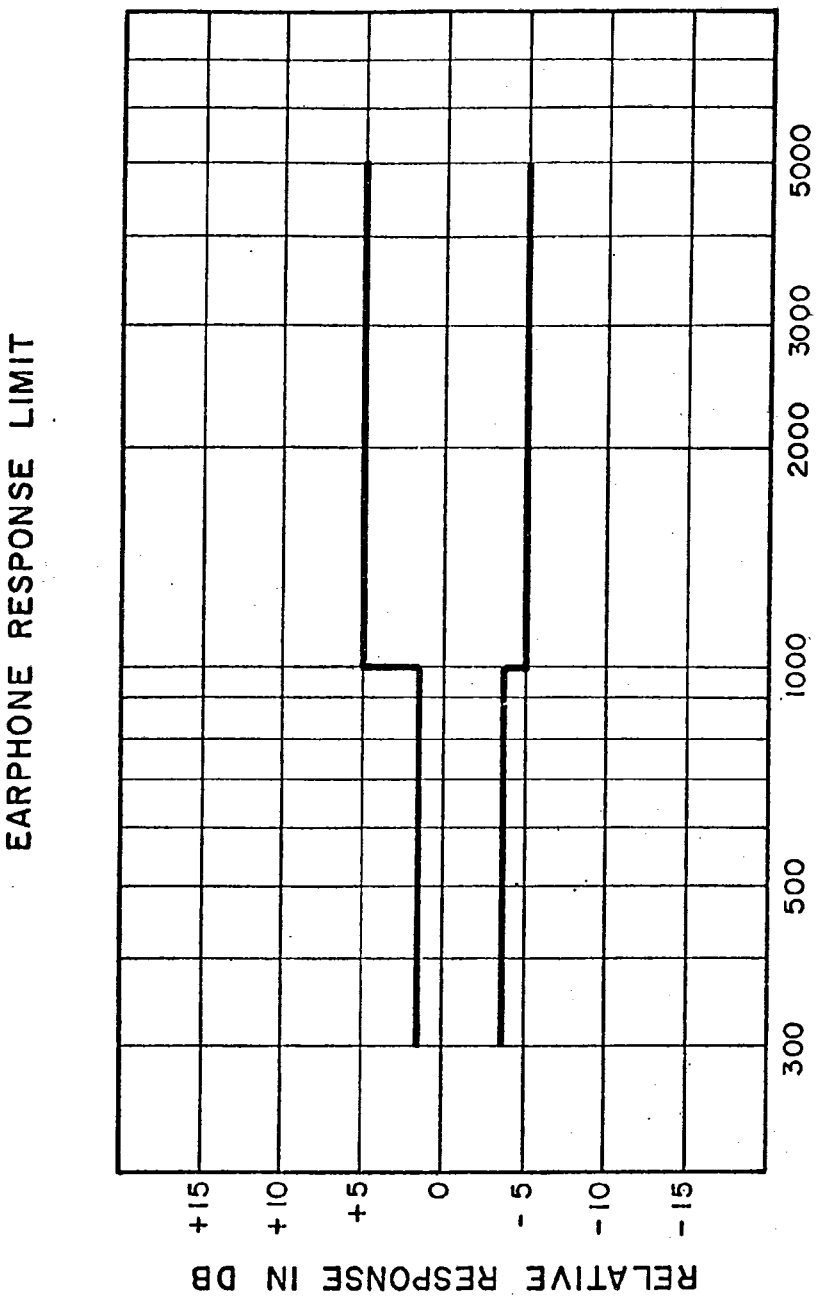


FIGURE 2

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TABLE I

Frequency (in Hz)	Deviation from 1000 Hz (in dB) Min.	Max.
300 - 1000	-3.5	+1.5
1000 - 5000	-5	+5

3.5.2 Distortion. The acoustic output of the earphone shall have no more than 5% total harmonic distortion over the audio frequency range of 300 to 5000 Hz, when measured as specified in 4.8.3.

3.5.3 Overload. The earphone shall show no more than 3 dB change from its original response curve, after being subjected to the test specified in 4.8.4.

3.5.4 Dielectric Strength and Insulation Resistance. There shall be no evidence of insulation breakdown when the earphone is subjected to a voltage of 100 VAC 60 Hz for ten (10) seconds applied between the terminals of the earphone and the insulated outer metal parts of the earphone. There shall be no decrease in insulation resistance below 1 megohm following the above test. See 4.8.5.)

3.5.5 Impedance. The impedance of the earphone at 1000 Hz shall be 17 ohms, plus or minus 2 ohms, as determined by the test of 4.8.2.

3.6 Service Conditions. The equipment shall meet the following service requirements:

3.6.1 Temperature. The equipment shall operate at ambient air temperatures ranging from -40°F to 150°F and shall withstand storage at temperatures from -70°F to 160°F. (See 4.13.1 and 4.13.2.)

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3.6.2 Humidity. The equipment shall be operable without degradation in specified performance and shall sustain no physical damage during and after prolonged exposure to relative humidities as high as 100 percent at all ambient air temperatures to 80°F, high absolute humidity corresponding to a dew point of 86°F occurring at an ambient temperature of 100°F, and low relative humidity of 5 percent at 125°F. (See 4.13.3.)

3.6.3 Altitude. The equipment shall operate without degradation in specified performance at altitudes up to 15,000 feet above sea level and shall withstand air transport at 50,000 feet. (See 4.13.4.)

3.6.4 Immersion. The equipment shall be immersionproof to a covering depth of three (3) feet of water for a period of not less than two (2) hours. (See 4.13.5.)

3.6.5 Salt Fog. The equipment shall be resistant to the corrosive effects of salt-sea atmosphere. (See 4.13.7.)

3.6.6 Blast (see 4.13.9). Thirty (30) rounds of blast at a peak pressure of 9.5 pounds per square inch, with no more than 3 dB degradation in performance.

3.6.7 Sand and Dust. The equipment shall withstand exposure to fine sand and dust particles, with wind speeds of 35 knots. (See 4.13.6.)

3.6.8 Fungus. The equipment shall be resistant to fungus growth as encountered in tropical climates. (See 4.13.8.)

3.6.9 Shock and Vibration. The equipment shall withstand shock and vibration induced during service usage and rough handling. (See 4.13.10 and 4.13.11.)

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3.7 CABEL ASSEMBLY (MICROPHONE) AND CORD ASSEMBLY KIT.

3.7.1 Construction (general).

3.7.1.1 Conductors. The nominal area of the conductor shall be 320 circular mils and the minimum area 310 circular mils. Each conductor shall consist of bunch stranded AWG No. 40 (0.003145 inch diameter) tinned cadmium bronze wire. The strands shall be twisted to have a right hand lay of approximately one-half inch. The individual strands, before bunching and plating, shall have a minimum tensile strength of 90,000 pounds per square inch and have a conductivity grade of at least 80 percent.

3.7.1.1.1 Conductor Insulation. The insulation shall consist of extruded teflon, not less than 0.006 inches thick. The insulation shall withstand a potential of 1000 volts commercial line frequency when applied in accordance with MIL-I-4997.

3.7.1.1.2 Tin-Coating. Each strand shall be tin-coated with commercially pure tin. The tin-coating, continuity, adherence, and finish shall meet the requirements of ASTM Standard B33-63.

3.7.1.1.3 Conductor Color Code. For circuit identification, the insulation for the six-conductor cord assembly shall be color coded as follows:

white: for the headset high conductor of each earphone

black: for the headset low conductor of each earphone

red: for the microphone high conductor

green: for the microphone low conductor

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The six-conductor cord shall be connected to the four-pin plug, U-174/U, as follows:

The two white conductors shall be connected to pin 2.

The two black conductors shall be connected to pin 4.

The red conductor shall be connected to pin 3.

The green conductor shall be connected to pin 1.

3.7.1.2 Stay Cords. The stay cords shall be composed of extra strength rayon threads braided together; they shall not abrade or otherwise damage the insulation of the conductors when the cord assembly is bent or stretched.

3.7.1.3 Jacket. The cord shall be covered with a braided jacket composed of black nylon, 210 denier, 3 ends, 20 carrier, approximately 30 picks per inch, in accordance with MIL-C-55668(EL).

3.7.2 Physical Characteristics.

3.7.2.1 Temperature Range. The cord shall be flexible and resilient throughout the temperature range of minus 55°C to plus 85°C and shall show no evidence of cracking or other damage as a result of exposure to this temperature range. (See 4.9.1.1)

3.7.2.2 Flexing Life (See 4.9.1.2.1). The six-conductor cord shall have a mean flex life of 300,000 flexes without showing evidence of:

a. damage to the outside jacket;

b. damage to the insulation of the individual conductors;

c. electrical discontinuity.

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3.7.3 Electrical Characteristics.

3.7.3.1 Dielectric Strength. For each finished cord assembly, the insulation shall withstand, without flash over or breakdown, the application of 500 VAC, 60 Hz for five (5) seconds. (See 4.9.1.3)

3.7.3.2 Insulation Resistance. The insulation resistance of the cord shall be not less than 100 megohms, when measured at or corrected to 15.5°C. (See 4.9.1.4)

3.7.4 Electromagnetic Compatibility.

3.7.4.1 Effect of External Magnetic Field. The sensitivity of the MK-896A/AIC to an external 400 Hz magnetic field shall not be more than 25% of that of a coil consisting of 123 turns of No. 34 enameled wire, closely wound in a single-layer solenoid upon a non-conductive, non-magnetic mandrel, 0.500 inch in diameter, placed in the same magnetic field. (See 4.9.1.5.1, 4.9.1.5.1.1, 4.9.1.5.1.2, and 4.9.1.5.1.3)

3.7.4.2 Stray Magnetic Field of the MK-896A/AIC. The stray magnetic field of the MK-896A/AIC shall cause no more than a five (5) degree deflection of a magnetic compass at a distance of 12 inches. (See 4.9.1.5.2)

3.7.4.3 Electromagnetic Interference (EMI). The MK-896A/AIC shall comply with the following emission and susceptibility requirements of Notice 4, MIL-STD-461A. (See 4.9.1.5.3)

CE03	RE02.1	RS01	RS03
CE05	RE02		RS03.1

(RE02.1 and RS03/RS03.1 shall be performed from 14 KHz to 1GHz.)

3.7.5 Conductor Strain Relief. The cable shall be anchored at the plug connectors in such a way that the cable can withstand the strain relief test specified herein. (See 4.9.1.6)

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3.7.6 Microphone Cord. The cord shall be WD-34/U in accordance with MIL-C-5898, or cord WD-27A/U in accordance with MIL-C-10392 may be used upon approval of the contracting officer.

3.8 Calibration System. The contractor shall establish and maintain a calibration system per MIL-C-45662 to control the accuracy of the measuring and test equipment used in testing of equipment procured under this specification.

3.9 Operational Requirement. If any of the electronic components of the MK-896A/AIC are procured as separate items, each item will be installed in a Government Furnished Equipment (GFE) MK-896A/AIC and shall be tested as indicated in 4.10.

3.10 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment (see 4.12). Individual items shall not be handpicked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristics, etc.

3.11 Preconditioning. The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6 (also see 4.5).

3.12 Workmanship. The equipment shall be manufactured and assembled in accordance wtih Requirement 9 of MIL-STD-454.

3.13 Systems Safety Engineering.

3.13.1 Personnel Hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirement 1, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly process, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual inspection (see 4.11).

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3.13.2 Edge Rounding Exposed. Exposed edges shall be rounded to a minimum radius of 0.04 inch (1 mm), and exposed corners to a minimum of 0.5 inch (13 mm).

3.13.3 Radioactive Materials. Radioactive materials shall not be used (e.g. luminous dials/markings, electron tubes, surge arrestors and lenses).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facility or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspection is deemed necessary to assure the supplies and services conform to the prescribed requirements.

4.2 Classification of Inspection. Inspection shall be classified as follows:

a. First Article inspection (does not include packaging). (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before packaging. (See 4.5)

(2) Quality conformance inspection of packaging (See 4.15)

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4.3 First Article Inspection. This inspection will be performed by the contractor and witnessed by a Government representative unless otherwise specified in the contract. It shall consist of Group A, B, and C inspections (see Tables II, III, and IV, respectively) on 25 samples. The first article samples will be inspected in this order: Group A, Group B, and divided up for Group C inspection (see Table IV) as follows:

Twenty units will be tested in accordance with test sequence of Table VII.

Five unassembled cords will be tested IAW 4.9, including all subparagraphs thereof.

4.4 Inspection Covered by Subsidiary Documents. The following shall be inspected under the applicable documents as part of the inspection of equipment before preparation for delivery:

Item	Where Required
Marking	3.2.2
Cord WD-34/U	3.7.6
Workmanship	3.12

4.5 Quality Conformance Inspection of Equipment Before Packaging. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection records. Each unit which will be subjected to Group A inspection, except first article samples, shall be preconditioned after final assembly (see 3.12).

4.5.1 Group A Inspection. This inspection, including sampling, shall conform to Table II and the ordinary inspection procedures of MIL-STD-105.

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4.5.1.1 Order of Inspection with Group A. Group A inspection shall be performed in any order which is satisfactory to the Government.

TABLE II

Group A Inspection

Inspection	Requirement Paragraph	Inspection Paragraph	AQL Major	AQL Minor
Visual and Mechanical	3.12	4.11	1.0%	4.0%
Microphone	3.13	4.11		
Response	3.4.1	4.7.1)	1.0%	*
Distortion	3.4.2	4.7.2)	for the	
Signal-to-Noise Ratio	3.4.3	4.7.3)	group	
Earphone				
Response	3.5.1	4.8.1)	1.0%	*
Distortion	3.5.2	4.8.3)	for the	
		)	group	
Headset-Microphone Kit				
Operational	3.9	4.10	1.0%	*

* All electrical and operational defects are considered major.

4.5.2 Group B Inspection. This inspection, including sampling, shall conform to Table III and to the special prcce-dures for small sample inspection of MIL-STD-105. Group B inspection shall be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

4.5.2.1 Order of Inspection Within Group B. Group B inspection shall be performed in any order which is satisfactory to the Government.

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TABLE III

Group B Inspection

Inspection	Requirement Paragraph	Inspection Paragraph	AQL
Microphone			
Dielectric Strength	3.4.4	4.7.4	6.5%*
Impedance	3.4.5	4.7.5	6.5%
Earphone			
Overload	3.5.3	4.8.4	6.5%*
Impedance	3.5.5	4.8.2	6.5%*
Dielectric Strength	3.5.4	4.8.5	6.5%*
Headset-Microphone Kit			
Interchangeability	3.10	4.12	6.5%*

* Normal inspection level shall be S-4.

4.5.3 Group C Inspection. This inspection shall consist of the inspection specified in Table IV and shall be performed on sample units that have been subjected to and meet Group A and B inspection.

4.5.3.1 Sampling for Inspection of Equipment. Twelve (12) samples per every 1,000 units produced shall be randomly selected by the Government. The contractor shall submit three (3) samples for each unit to be tested as specified in Table VII.

4.5.3.2 Noncompliance. If a sample unit fails Group C inspection evaluation, the contractor shall immediately investigate the cause of failure and report to the Government inspector the results thereof, details of the corrective action taken on the process, and all units of product which were manufactured with the same conditions, materials, processes, etc.

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If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

TABLE IV

Group C Inspection

Inspection	Requirement Paragraph	Inspection Paragraph
Headset-Microphone Kit		
Temperature	3.6.1	4.13.1, 4.13.2
Humidity	3.6.2	4.13.3
Altitude	3.6.3	4.13.4
Immersion	3.6.4	4.13.5
Salt Spray	3.6.5	4.13.7
Blast	3.6.6	4.13.9
Sand and Dust	3.6.7	4.13.6
Fungus	3.6.8	4.13.8
Shock and Vibration	3.6.9	4.13.11, 4.13.10
Effect of External Magnetic Field	3.7.4.1	4.9.1.5.1, 4.9.1.5.1.1, 4.9.1.5.1.2, 4.9.1.5.1.3
Stray Magnetic Field of the MK-896A/AIC	3.7.4.2	4.9.1.5.2
Electromagnetic Interference (EMI)	3.7.4.3	4.9.1.5.3
Cord Assembly		
Temperature Range	3.7.2.1	4.9.1.1
Flexing Life	3.7.2.2	4.9.1.2.1
Dielectric Strength	3.7.3.1	4.9.1.3
Insulation Resistance	3.7.3.2	4.9.1.4
Conductor Strain Relief	3.7.5	4.9.1.6

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4.5.4 Reinspection of Conforming Group B and C Sample Units. Unless otherwise specified, sample units which have been subjected to and passed Group B or C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass Group A inspection after repair of all damages.

4.6 Bounce Preconditioning. The unit shall be placed in its normal operation position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of three (3) inches as the unit bounces. The unit shall be subjected to this preconditioning for one (1) minute. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to quality conformance inspection.

4.7 Microphone Tests.

4.7.1 Response Test. Position the microphone, under test, with its face ¼ inch on axis to the driver unit of an artificial voice (Bruel & Kjaer type 4219, or equal) with the diaphragm of the microphone parallel to the driver unit. The artificial voice shall have been previously calibrated by means of a condenser microphone (Bruel & Kjaer type 4133, or equal, certified traceable to National Bureau of Standards). This calibration will be conducted with an applied constant sound pressure of 3.6 pascals per square centimeter (105dB sound pressure level, Re 20 micropascals). Determine the microphone output readings using a Hewlett-Packard 334A distortion analyzer or equal, set for voltage measurement (or a suitable A.C. electronic voltmeter having an impedance of at least ½ megohm), connected across a non-inductive load resistance of five (5) ohms. The microphone shall meet the requirements of 3.4.1.

4.7.2 Distortion. Harmonic distortion shall be determined using the same equipment and circuitry as used in 4.7.1 except that the constant sound pressure shall be 125dB referenced at 20 micropascals and the output of the microphone terminated in five (5) ohms shall be connected to a Hewlett-Packard Distortion Analyzer Model 334A, or equal, for distortion measurements.

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The distortion shall not be greater that 6%. Measurements will be made from 700 Hz to 1000 Hz, inclusive, at 100 Hz increments and from 1000 to 5000 Hz, inclusive, at 500 Hz increments. The Microphone shall meet the requirements of 3.4.2.

4.7.3 Signal-to-Noise Ratio.

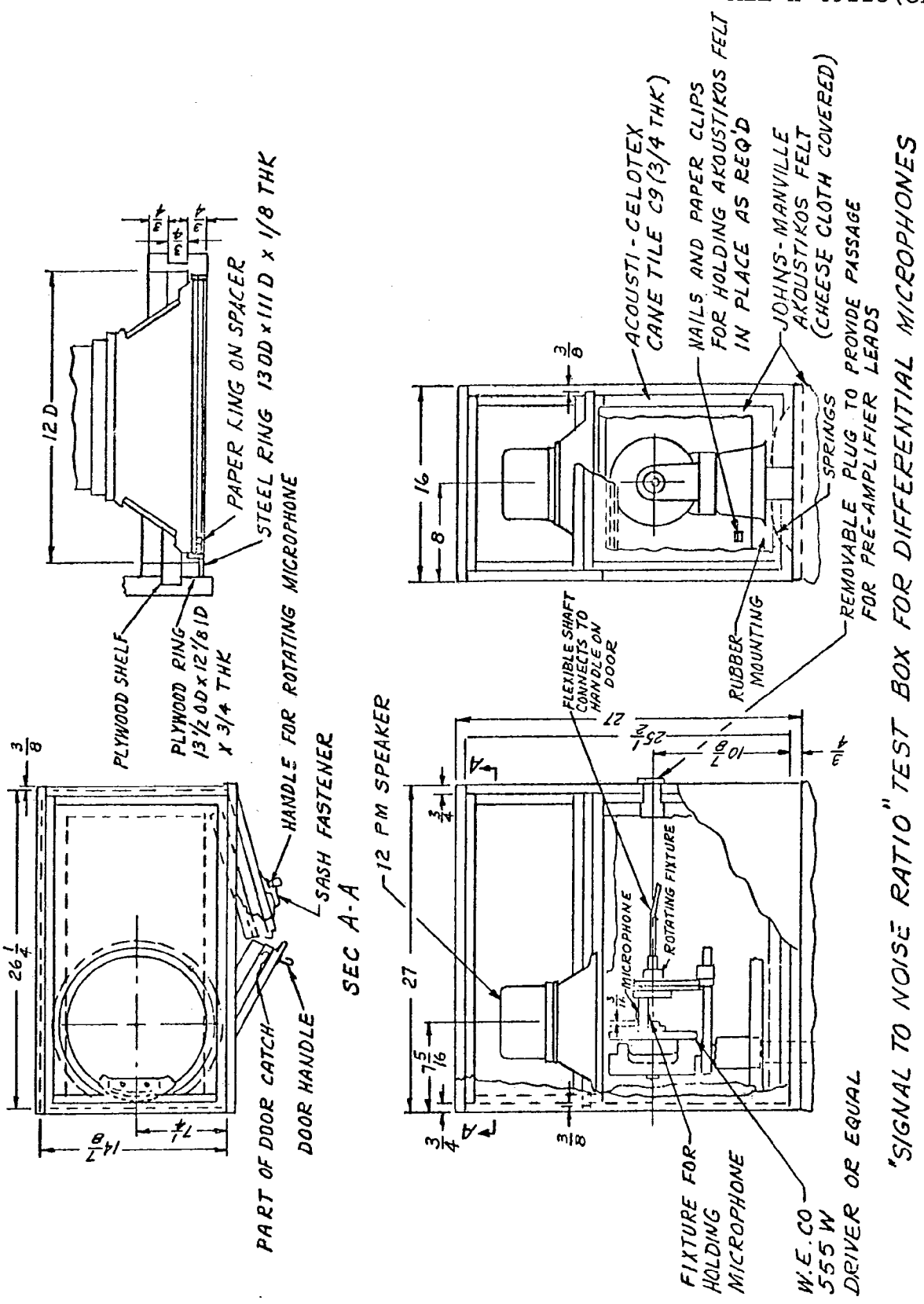
4.7.3.1 Noise Spectrum. By means of apparatus detailed in Figure 3, and the test circuit of Figure 4, provision shall be made for the production of the noise spectrum, Table V, at an rms sound pressure of 115dB above a referenced level of 20 micropascals as measured at the microphone diaphragm. The "Noise" source shall incorporate a loudspeaker capable of producing a sound pressure of 115dB above a reference level of 20 micropascals at the rear, as well as the front of the microphone. The loudspeaker shall be located as far away from the microphone as practicable. The calibrating condenser microphone shall be mounted in the test rig, in the box, 3/16 inch from the mouth of the driver unit (signal source), which shall be short-circuited during adjustments of the noise spectrum. The microphone shall meet the requirements of 3.4.3.

TABLE V

Noise Spectrum

Frequency in Hz	Sound Pressure in dB Relative to Sound Pressure at 130 Hz
40	0
70	0
130	0
300	-5
600	-9
1000	-13
2000	-17
3000	-19
4000	-21

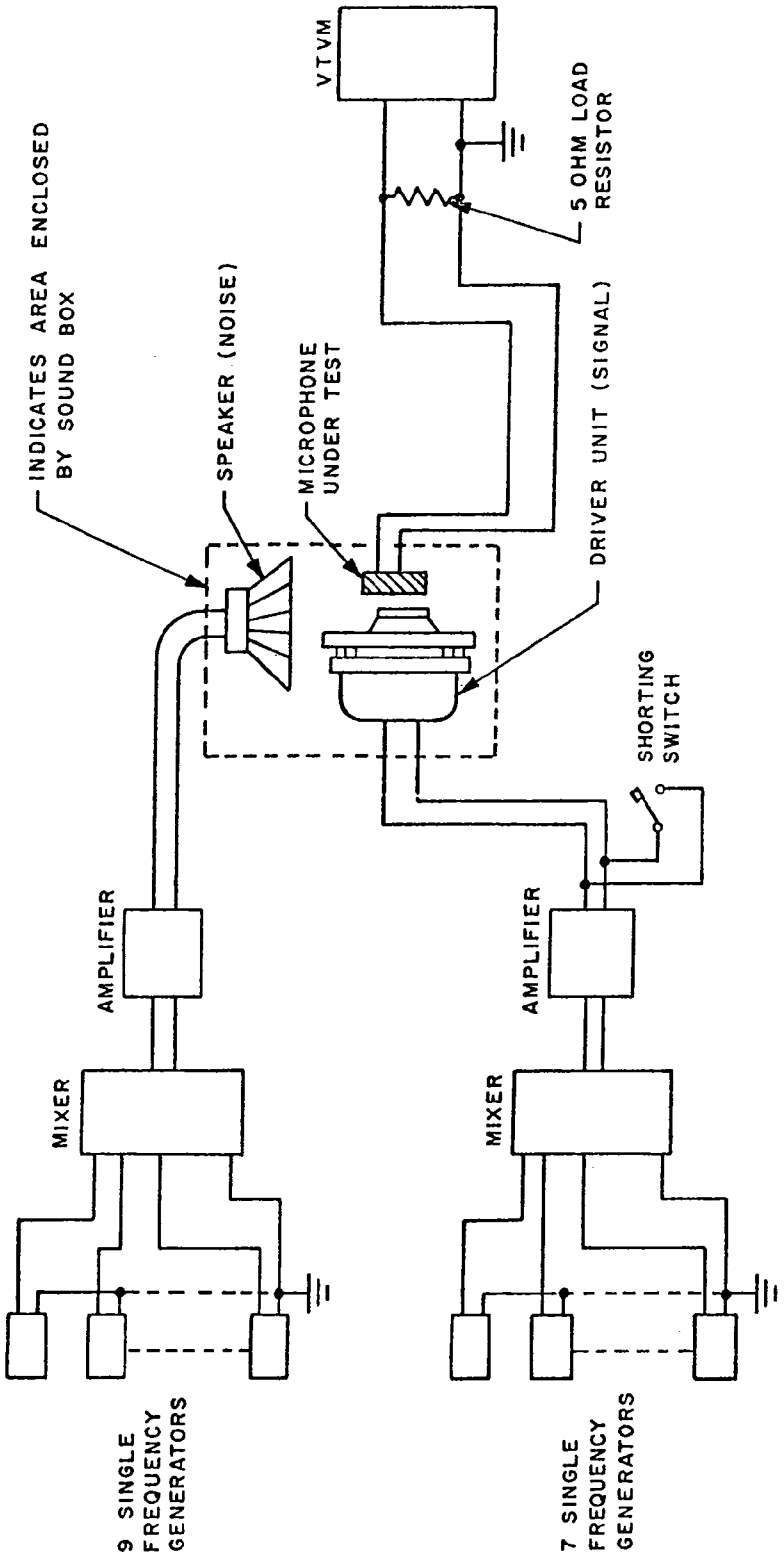
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'SIGNAL TO NOISE RATIO" TEST BOX FOR DIFFERENTIAL MICROPHONES

FIGURE 3

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MICROPHONE SIGNAL-TO-NOISE RATIO TEST CIRCUIT

FIGURE 4

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4.7.3.2 Signal Spectrum. By means of the apparatus as set up previously, provision shall also be made for the production of the following signal spectrum, Table VI at an rms sound pressure of 115dB above a reference level of 20 micropascals. The "signal" source shall be the driver unit. The calibrating condenser microphone shall be positioned 3/16 inch directly in front of the driver unit.

TABLE VI

Signal Spectrum

Frequency in Hz	Sound Pressure in dB Relative to Sound Pressure at 130 Hz
130	0
300	+7
600	+8
1000	+5
2000	-3
3000	-7
4000	-9

4.7.3.3 Measurement of the Signal-to-Noise Ratio. The microphone shall be mounted in its test rig with its face 3/16 inch from the mouth of the driver unit. The microphone shall be connected to the test circuit shown in Figure 4. The noise spectrum shall be applied for three (3) seconds, and the output of the microphone shall be measured. The noise spectrum shall be removed, and the signal spectrum shall be applied for three (3) seconds. The output of the microphone shall be measured. The dB difference between the "signal" value and the "noise" value is the signal-to-noise ratio, and shall be in accordance with 3.4.3.

4.7.4 Dielectric Strength and Insulation Resistance Test. The microphone shall be tested for compliance with 3.4.4.

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4.7.5 Impedance. The microphone shall be placed in front of the driver unit as specified in 4.7.1 with a calibrated variable resistor switch in the open circuit position, replacing the 5 ohm resistor in Figure 4, and the electronic voltmeter connected across the terminals of the microphone. Increase the sound pressure of the driver unit at 1000 Hz until the output of the microphone is 0.0001 volt. Introduce resistance across the microphone until the output is reduced to 0.00005 volt. The resistance thus determined shall be taken as the electrical impedance of the microphone and shall comply with 3.4.5.

4.8 Earphone Tests.

4.8.1 Earphone Response Test. Available power frequency response measurements shall be made starting at 300 Hz and extending through 5000 Hz, in sufficient detail to establish definitely the shape of the curve. Response measurements shall be made in accordance with "Coupler Calibration of Earphones," "ANSI Standard Z24.9-1949." The contractor shall supply a 6 cc coupler. The outside dimensions of the coupler shall be such as to provide the proper seating and sealing of the earphone.

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4.8.1.1 Testing Procedure. The earphone element shall be mounted as shown in the test circuit (see Figure 5). Adjust the output from the audio oscillator to supply a specified power input (see 3.5.1) to the element at 1000 Hz as measured by the vacuum tube voltmeter (VTVM) #1. The frequency of the input shall be varied continuously from 300 Hz to 5000 Hz, inclusive. The acoustic output shall be recorded on a direct-writing strip graph chart using an automatic plotter or curve tracer with a minimum writing speed of ten (10) inches per second and a maximum chart speed of thirty (30) inches per minute. The output shall be recorded in the dB value above 20 micropascals. The writing speed and chart speed shall be noted. In lieu of this data recording method, and at the option of the manufacturer, point-to-point measurements may be made every 100 Hz from the lowest end of the frequency range to 1000 Hz, and every 250 Hz from 1000 Hz to 5000 Hz. These measurements shall be of the acoustic output from the calibrated microphone and preamplifier unit as indicated by the reading of VTVM #2, and converted to the equivalent dB value above 20 micropascals, using the most recent available calibration curve for the test microphone. The data shall be recorded and a graph drawn.

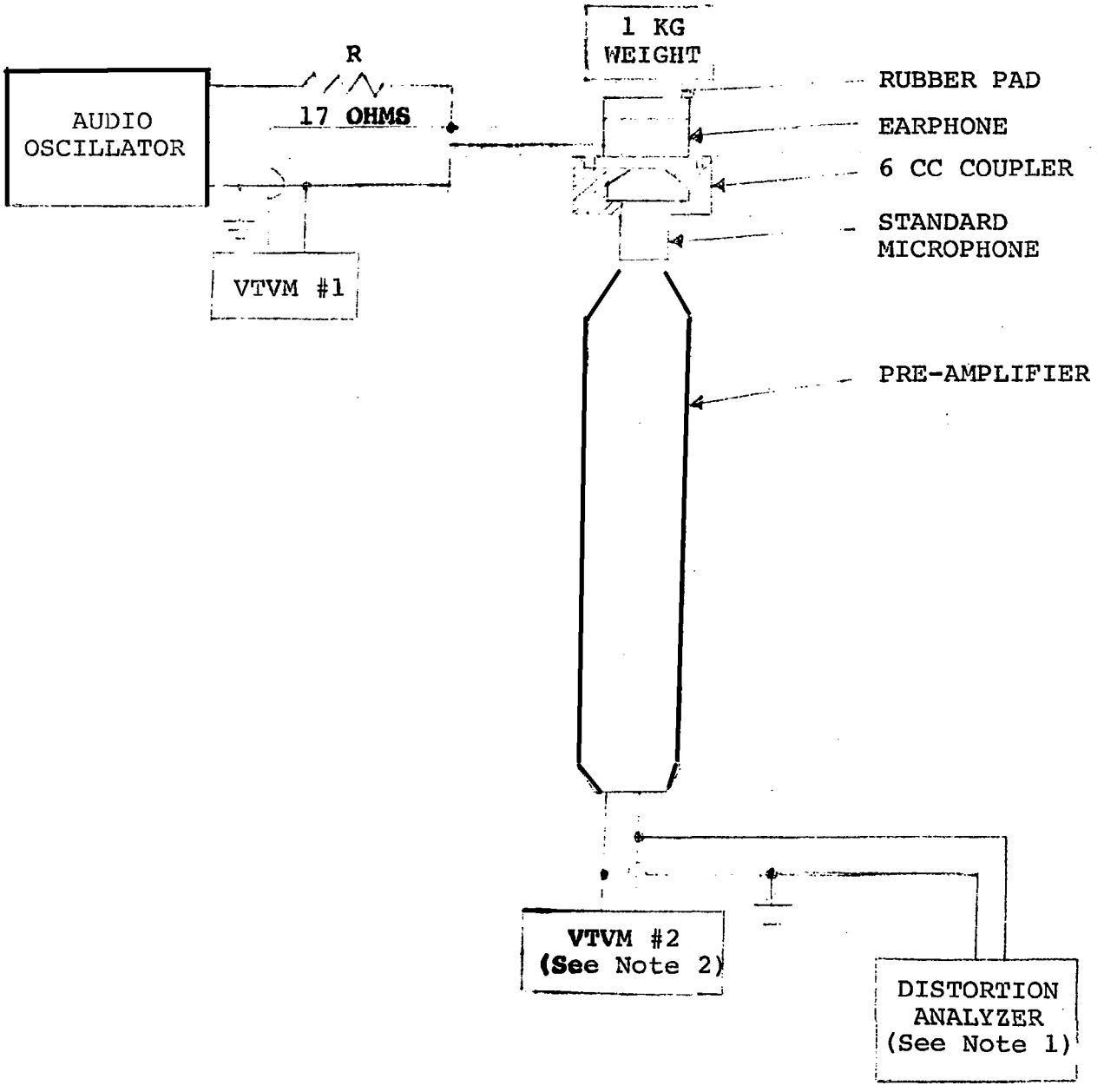
4.8.1.2 Test Equipment. The test equipment used for the response test shall meet the following requirements:

a. Calibrating Microphone. A Bruel & Kjaer Instruments Inc. Model 4160 (Western Electric Co. 640AA equivalent), one-inch, pressure-type microphone, or equal, shall be used for measuring sound pressure. It should be calibrated by the reciprocity method in accordance with ANSI S1.10-1966 USA Standard Method for the Calibration of Microphones.

b. Audio Oscillator. The audio oscillator shall have a frequency range of at least 100 to 10,000 Hz, shall have a high degree of stability in both output voltage and frequency, and shall have a wave form distortion of less than 2 percent.

c. Vacuum-Tube Voltmeter. The vacuum-tube voltmeters used must have flat frequency response (±1dB) from at least 100 to 10,000 Hz and must be capable of measuring voltage from 0.0005 volt rms to 10 volts or more.

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Note 1: DISTORTION ANALYZER REQUIRED FOR DISTORTION TEST ONLY.

Note 2: VTVM OR AUTOMATIC RECORDER

EARPHONE FREQUENCY RESPONSE TEST CIRCUIT

FIGURE 5

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d. Microphone Pre-Amplifier. The microphone pre-amplifier shall have a flat response (±1dB) over a frequency range of at least 100 to 10,000 Hz, shall have a high degree of stability, and shall have distortion of less than 2 percent.

4.8.2 Impedance. The impedance of the earphone shall be measured with 0.13 volt at 1000 Hz applied to the earphone terminals. The impedance shall be determined either by measuring the voltage across and the current through the earphone, or with an impedance bridge. The impedance of the earphone when mounted on the coupler shall meet the requirements of 3.5.5.

4.8.3 Distortion Test. Harmonic distortion shall be determined using the same equipment and circuitry as used in 4.8.1.1 except that the constant sound pressure shall be 110dB referenced at 20 micropascals and the acoustic output from the calibrated microphone and preamplifier unit shall be connected to a Hewlett-Packard distortion analyzer Model 334A, or equal, for distortion measurements. The total harmonic distortion shall not be greater than 5 percent over the audio frequency range of 300 Hz to 5000 Hz. Total harmonic distortion shall comply with the requirements of 3.5.2.

4.8.4 Overload. After operation of the earphone for 8 hours with 300 milliwatts input power (2.3 volts) at 1000 Hz, the response shall be tested per 4.8.1 to establish compliance with the requirements of 3.5.3.

4.8.5 Dielectric Strength and Insulation Resistance Test. The earphone shall be tested for compliance with 3.5.4.

4.9 Cord Test.

4.9.1 Cord Assembly Tests.

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4.9.1.1 Temperature Range. The unterminated cord shall be placed within a test chamber wherein a temperature of plus 85°C (plus 185°F) is maintained for a period of four (4) hours. The cord shall be wrapped about a mandrel while at this temperature without damage to the cord jacket or conductor insulation. The diameter of the mandrel shall be the same as the nominal cable diameter. The cord shall be attached to the mandrel and suspended vertically with the lower end weighted sufficiently to keep the specimen taut and to permit wrapping without handling. The cord shall be inspected for damage or deformation and then subjected to a test chamber wherein a temperature of minus 55°C (-67°F) is maintained. After twenty (20) hours of storage at -55°C, the cord shall be wrapped about the mandrel, while at this temperature, in the same manner as it was at the positive extreme temperature. The cord shall be examined for cracks or any other damage to the jacket or conductor insulation. The cord shall meet the requirements of 3.7.2.1.

4.9.1.2 Flexing Life.

4.9.1.2.1 Six-Conductor Cord. Samples of the six-conductor unterminated cord shall be suspended through holes in a metal bar having a cross section of one (1) inch by one (1) inch. The cord shall be secured in such a manner that it will not turn in the hole during the flexing test. The diameter of the through holes shall be 0.005 to 0.010 inch greater than the diameter of the cord which is undergoing test. The through hole shall incorporate a 1/8-inch radius at the end from which the cord emerges (bending point). A knot shall be tied near the free end of the cord, and a one-pound weight shall be hung from the knot to maintain the cord in a vertical position. The cord assembly shall be suspended through the test bar in a manner to prevent flexing between the top of the one-inch bar and the attachment point of the stay cord. The test apparatus shall provide an electrical potential of 6 VDC for each conductor under test. A control circuit shall be provided that will positively detect a current interruption of five (5) milliseconds or less to prevent broken conductor from flexing with ends touching without detection. The metal bar, supported horizontally, shall then be rotated about its logitudinal axis back and forth through an angle of 120 degrees (60 degrees each side of vertical) at a rate of approximately 85 cycles per minute. The arithmetical mean of the flexing cycles of the sample of the cord, before electrical discontinuity occurs, shall be taken as the flexing life of the cord. The cord shall meet the requirements of 3.7.2.2.

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4.9.1.2.1.1 Tin-Coating. The tin-coating on the conductor strands and the continuity, adherence, and finish shall pass the applicable tests specified in ASTM Standard B33-63. (See 3.7.1.1.2.)

4.9.1.3 Dielectric Strength. The cord assembly shall be subjected to an alternating potential as outlined in 3.7.3.1. The voltage source shall have a capacity of not less than 3 kVa.

4.9.1.4 Insulation Resistance. Immediately after the cord has successfully withstood the dielectric strength test, the insulation shall be measured. The insulation resistance shall meet the requirements of 3.7.3.2. If the measurement is made at any other temperature, the contractor shall correct the measured value of insulation resistance to 15.5°C. The contractor shall demonstrate that the correction factor is accurate for his compound. In making the insulation resistance test, the test may be terminated in less than one minute if the galvano-meter has ceased fluctuating and the reading indicates that a steady insulation resistance value has been obtained.

4.9.1.5 Electromagnetic Interference (EMI).

4.9.1.5.1 Effect of External Magnetic Field. The sensitivity of the MK-896A/AIC to an external A.C. magnetic field shall be measured by comparing the induced open-circuit voltage across the terminals of the microphone and the earphones when placed in a 400 Hz magnetic field with the induced open-circuit voltage across the terminals of a calibrating coil of known sensitivity placed in the same magnetic field.

4.9.1.5.1.1 Field Coil. The magnetic field for this test may be produced by a coil of about 700 turns of No. 18 enameled wire. The coil should be six (6) inches in cross section or six (6) inches in diameter and winding-length of ten (10) inches.

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4.9.1.5.1.2 Calibrating Coil. The calibrating coil shall consist of a single-layer solenoid wound with 123 turns of No. 34 enameled wire on a non-conductive, non-magnetic mandrel, 0.500 inch in diameter.

4.9.1.5.1.3 Measurement Procedure. Excite the field coil with about 250 milliamperes of 400 Hz current. Connect the microphone and earphones to an electronic voltmeter. Closely twist the leads from the microphone and earphones to the voltmeter. Orient the microphone and earphones in the center of the field coil to maximum output voltage and record this value. Then substitute the calibrating coil for the microphone and earphones. Orient the calibrating coil to maximum output voltage in that portion of the field coil previously occupied by the microphone and earphones. Record the output voltage. MK-896A/AIC shall meet the requirements of 3.7.4.1.

4.9.1.5.2 Stray Magnetic Field of the MK-986A/AIC. The stray magnetic field of the MK-896A/AIC shall be determined, as follows, using a Keuffel & Esser type No. 5600 compass or its approved equivalent. The microphone and earphone transducers shall be placed with their geometric center twelve (12) inches from the pivot point of the compass needle and on the perpendicular bisector of the needle in the plane of rotation of the needle. The microphone and earphones shall be oriented in all directions and the maximum deflection of the compass observed. This test shall be made in a location substantially free from stray magnetic disturbances. MK-896A/AIC shall meet the requirements of 3.7.4.2.

4.9.1.5.3 Electromagnetic Interference (EMI). The MX-896 A/AIC shall comply with the requirements of 3.7.4.3 using the test methods and procedures of Notice 3(EL), MIL-STD-462.

4.9.1.6 Conductor Strain Relief. (See 3.7.5)

4.9.1.6.1 Pull Test at Cable Clamp. A pull of 30 pounds shall be applied for a minimum of two (2) minutes to the cable in the direction of the axis of the cable as it passes through the cable clamp mounted on the headset.

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4.9.1.6.2 Pull Test at Telephone Plug. A pull of 25 pounds shall be applied for a minimum of two (2) minutes to the cable in the direction of the axis of Telephone Plug U-174/U.

4.9.1.6.3 Pull Test at Plug Connector. A pull of eight (8) pounds shall be applied for a minimum of thirty (30) minutes to the cable in the direction of the axis of Plug Connector U-172/U and U-173/U.

4.10 Operational Test. Prior to acceptance, each completely assembled unit shall be tested by a talk test to insure correct wiring and satisfactory operation. (See 3.9.)

4.11 Visual and Mechanical Inspection. The equipment shall be examined for the defects listed in MIL-STD-252. (See 3.12 and 3.13.) An inspection shall be performed to verify compliance with those portions of 3.13 which can be determined visually.

4.12 Inspection for Dimensional Interchangeability. Each replaceable part listed below in the selected Transducer Equipments shall be interchanged with the corresponding part in the approved preproduction sample in sequential order. At the completion of this inspection, the interchanged parts shall be reassembled in their original transducer equipments. Non-interchangeability of these parts constitute failure. (See 3.10.)

Microphone element
Microphone boom
Earphone elements

4.13 Service Conditions Tests. The service conditions tests shall conform to the test sequence of Table VII.

4.13.1 High Temperature. The equipment shall be subjected to the test of Method 501, Procedure I, of MIL-STD-810. In step 4, the operating temperature shall be 65.5°C (150°F). The test of 4.10 shall be performed in Step 5. In Step 7, the equipment shall meet the requirements of 3.4.1 and 3.5.1 with degradation not to exceed 3dB. (See 3.6.1.)

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4.13.2 Low Temperature. The equipment shall be subjected to the test of Method 502, Procedure I, of MIL-STD-810. The storage temperature (Step 2) shall be -57°C (-70°F) and shall be maintained for a period of not less than two (2) hours following stabilization of the test item. Low operating temperature (Step 4) shall be -40°C (-40°F). The test of 4.10 shall be performed in Step 5. In Step 7, the equipment shall meet the requirements of 3.4.1 and 3.5.1 with degradation not to exceed 3dB. (See 3.6.1.)

4.13.3 Humidity. The equipment shall be subjected to the test of Method 507, Procedure II, of MIL-STD-810. In Step 6, the test of 4.10 shall be performed during the last five (5) hours of the last cycle. In Step 8, the equipment shall meet the requirements of 3.4.1 and 3.5.1 with degradation not to exceed 3dB. (See 3.6.2.)

4.13.4 Altitude. (See 3.6.3.)

4.13.4.1 Altitute Test (Operating). The microphone and earphone units shall be placed in an altitude chamber. The response of the microphone and earphone units at ground level shall be obtained at 30C, 600, 1000, 2000, and 5000 Hz. Pressure inside the chamber shall then be reduced to that corresponding to an altitude of 15,000 feet. The response of the microphone and earphone units shall be taken again at the same frequencies as mentioned above. The degradation in response of either unit shall be not more than 5dB from the requirements of 3.4.1 and 3.5.1.

4.13.4.2 Altitude Test (Non-Operating). The microphone and earphone shall be subjected to five (5) varying pressure cycles. Each pressure cycle shall consist of thirty (30) minutes at 3.4 inches of mercury (approximately 50,000 feet), and thirty (30) minutes at thirty (30) inches of mercury (approximately sea level). The pressure transition shall be approximately 5000 feet per minute. After the fifth cycle, the response characteristics of the earphone and microphone shall be measured and shall not exhibit a degradation in performance in excess of 5dB from the requirements of 3.4.1 and 3.5.1.

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4.13.5 Immersion. The equipment shall be subjected to the test of Method 512, Procedure I, of MIL-STD-810. Following the immersion period, the equipment shall meet the requirements of 3.4.1 and 3.5.1 with no degradation in performance. This test shall be performed prior to and following vibration and shock testing of the same sample test item. (See 3.6.4.)

4.13.6 Sand and Dust. The equipment shall be subjected to the test of Method 510, Procedure I, of MIL-STD-810. In Step 5, the equipment shall meet the requirements of 3.4.1 and 3.5.1 with no degradation in performance. (See 3.6.7.)

4.13.7 Salt Fog. The equipment shall be subjected to the test of Method 509, Procedure I, of MIL-STD-810. Operation of the equipment shall not be required and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as any visible degradation of the equipment surfaces that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface. (See 3.6.5)

4.13.8 Fungus. The equipment shall be subjected to the test of Method 508, Procedure I, of MIL-STD-810. Operation of the equipment is not required, but examination at the conclusion of the exposure period shall reveal no evidence of viable fungus on any surfaces. (See 3.6.8.)

4.13.9 Blast Test. Each earphone and microphone element being tested shall be mounted on the carriage of the U.S. Navy Simulated Gun Blast Equipment, in accordance with Bureau of Ships Drawing SK-N-864, with the front edge of the earphone or microphone element in the test plane and with its axis coincident with that of the explosion chamber. The earphone and microphone element shall be subjected to thirty (30) rounds of blast at a peak pressure of 9.5 pounds per square inch. The earphone and microphone units shall not suffer a degradation in their response of more than 3dB from the performance values of 3.4.1 and 3.5.1. (See 3.6.6.)

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4.13.10 Vibration. The equipment shall be subjected to sinusoidal vibration over a frequency range of 5 to 20 Hz at a constant displacement of 0.10 inches, and over a frequency range of 20 to 500 Hz at a constant acceleration of 2 g's. One vibration cycle (5 to 500 to 5) shall consume fifteen (15) minutes and shall be at a linear rate such that the sweep time for the 5-20-5 range is six (6) minutes, the 20-110-20 range is five (5) minutes, and the 110-500-110 range is four (4) minutes. The test items shall be mounted in a horizontal position and cycled as above the vibration in a direction perpendicular to the plane of the earphone and microphone for a period of two (2) hours. Upon completion of the test, the equipment shall be undamaged and shall meet the requirements of 3.4.1 and 3.5.1 with degradation not to exceed 3dB. (See 3.6.9.)

4.13.11 Shock, Drop. The microphone and earphone elements shall be dropped at random from a height of six (6) feet, twelve (12) times, on a concrete floor. At completion of the test, the equipment shall meet the requirements of 3.4.1 and 3.5.1 with degradation not to exceed 3dB. There shall be no evidence of breaking, cracking, or physical damage. Minor chipping shall not be considered a failure. (See 3.6.9.)

TABLE VII

Test Sequence

	Unit 1	Unit 2	Unit 3	Unit 4
Altitude	1			
Low Temperature		1		
High Temperature			1	
Sand & Dust				1
Immersion				2 & 5
Humidity	2			
Vibration				3
Shock, Drop				4
Blast		2		
Salt Fog		3		
Fungus			2	

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4.14 Government Verification of Calibration System. All operations performed by the contractor in compliance with MIL-C-45662 will be subject to government verification at unscheduled intervals.

4.15 Quality Conformance Inspection for Packaging. Quality conformance inspection for packaging shall be in accordance with Group A, Group B, and when required, (see 6.2e), Group C (rough handling) requirements of MIL-P-116. Inspection lots shall be in accordance with MIL-P-116.

5. PACKAGING

5.1 Packaging Requirements. The requirements for packaging shall be in accordance with SPI 1G00135.

6. NOTES

6.1 Intended Use. The equipment covered in this specification is intended for use as audio accessories for regular and extra large Helmet, Flying Type SPH-4.

6.2 Ordering Data. Procurement documents should specify the following:

a. Construction (construction shall be in accordance with the contract per 3.1.1).

b. Title, number, and data of this specification and any amendment thereto.

c. Type required.

d. Level A and B preservation and packing. (See Section 5.)

e. When rough handling tests are required. (See Section 4.)

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f. Preproduction pack(s) as follows:

Makeup of pack(s).

Number of each kind of pack to be submitted.

Inspection to be performed thereon.

g. Marking and shipping of samples.

h. Place of final inspection.

6.3 Engineering Data. Headset-Microphone Kit MK-896A/AIC replaces the MK-896/AIC which was part of the obsolete APH-5 Aviator's Helmet. Both have been purged from the supply system.

6.4 Group C Inspection. Approval to ship may be withheld, at the discretion of the Government pending the decision from the contracting officer or the adequacy of corrective action.

Custodian:

ARMY-CR

Preparing Activity:

ARMY-CR

Project No. 5965-A177

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